2_10xPBMC_STARsolo_noPlatelet

August 31, 2026

[1]:
```
library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xPBMC")
dir.create("data_10xPBMC")

colorPBMC <- "#81B29A"

dataset_id <- "10xPBMC"
sample <- "pbmc8k"
```

```
Loading required package: SeuratObject

Loading required package: sp


'SeuratObject' was built under R 4.3.3 but the current version is
4.4.1; it is recomended that you reinstall 'SeuratObject' as the ABI
for R may have changed

'SeuratObject' was built with package 'Matrix' 1.6.4 but the current
version is 1.7.5; it is recomended that you reinstall 'SeuratObject' as
the ABI for 'Matrix' may have changed


Attaching package: 'SeuratObject'


The following objects are masked from 'package:base':

    intersect, t


Warning message:
"package 'dplyr' was built under R version 4.4.3"
```

```
Attaching package: ‘dplyr’


The following objects are masked from ‘package:stats’:

    filter, lag


The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union


Warning message:
"package ‘RColorBrewer’ was built under R version 4.4.3"
```

'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'

```
Warning message in dir.create("figures_10xPBMC"):
"'figures_10xPBMC' already exists"
Warning message in dir.create("data_10xPBMC"):
"'data_10xPBMC' already exists"
```

[2]:

```
# celltype annotation

celltypes <- read.table("whitelists/celltype_annotation_sub_min.tsv") # file␣
↪passed to snakemake

celltypes$V2 <- gsub(pattern="CD14pos_Monocytes", replacement =␣
↪"Monocytes_CD14pos", celltypes$V2)
celltypes$V2 <- gsub(pattern="FCGR3Apos_Monocytes", replacement =␣
↪"Monocytes_FCGR3Apos", celltypes$V2)
celltypes$V2 <- gsub(pattern="Naive_CD4_T", replacement = "T_CD4_Naive",␣
↪celltypes$V2)
celltypes$V2 <- gsub(pattern="Memory_CD4_T", replacement = "T_CD4_Memory",␣
↪celltypes$V2)
celltypes$V2 <- gsub(pattern="CD8A", replacement = "T_CD8", celltypes$V2)
celltypes$V2 <- gsub(pattern="Megakaryocytes", replacement = "Platelet",␣
↪celltypes$V2)

PBMCcelltypeColors <- c("B"="#91bcca",
                "Monocytes_CD14pos"="#3d405b",
                "T_CD8"="#f2cc8f",
                "Dendritic"="#c492a7",
                "Monocytes_FCGR3Apos"="#e78063",
               # "Platelet"="#c4b3ab",
                "T_CD4_Memory"="#81b29a",
```

```
                "T_CD4_Naive"="#4d7362",
                "NK"= "#88535a")
```

[15]:
```
path <- paste0("/mnt/volume_1p5T/results/STARsolo_EM_TE/",dataset_id,"/
 ↪",sample,"/TE_Solo.out/Gene/")

# get filtered barcodes
STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/
 ↪",dataset_id,"/",sample,"/best_Solo.out/Gene")
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1␣
 ↪# read barcodes seleced by STARsolo
# barcodes of platelets
plateletBarcodes <- celltypes[celltypes$V2=="Platelet","V1"]
filteredBarcodes <- setdiff(filteredBarcodes, plateletBarcodes)

STARsolo_TE_EM_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "/raw/UniqueAndMult-EM.mtx"),
        cells = paste0(path, "/raw/barcodes.tsv"),
        features = paste0(path, "/raw/features.tsv")
    ) # read matrix
STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only␣
 ↪barcodes passing thresholds

nCells <- length(filteredBarcodes)
thrMinCells <- round(nCells * 0.05)
thrMinCells

objTE_STARsolo <- Seurat::CreateSeuratObject(STARsolo_TE_EM_mat,
        project = "STARsolo_TE_EM",
        min.cells = thrMinCells, min.features = 50
    )


objTE_STARsolo
```

53

```
Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"
```

```
An object of class Seurat
4956 features across 1056 samples within 1 assay
Active assay: RNA (4956 features, 0 variable features)
 1 layer present: counts
```

[4]:
```
annotation_stellarscope <- read.table("annotation/annotation_stellarscope_hg38.
 ↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
head(annotation_stellarscope)
```

A data.frame: 6 × 10

	chr <chr>	start <int>	end <int>	sw_score <int>	strand <chr>	locusID <chr>	family <chr>	subfamily <chr>
1	chr1	67108754	67109046	1892	+	L1P5	L1	L1P5
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
3	chr1	25165804	25166380	4085	+	L1MB5	L1	L1MB5
4	chr1	33554186	33554483	2285	-	AluSc	Alu	AluSc
5	chr1	41942895	41943205	2451	-	AluY_dup1	Alu	AluY
6	chr1	50331337	50332274	1587	+	HAL1	L1	HAL1

[16]:

```r
# Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
starsoloTEs <- Features(objTE_STARsolo)

filteredStarsoloTEs <- annotation_stellarscope[annotation_stellarscope$stellarscopeID %in% starsoloTEs, ] %>%
    dplyr::filter(!class %in% classesToExclude) %>%
    pull(stellarscopeID)

print("Percentage of TEs in removed orders:")
print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) * 100)
```

```
[1] "Percentage of TEs in removed orders:"
[1] 0.02017756
```

[17]:

```r
objTE_STARsolo <- objTE_STARsolo[intersect(starsoloTEs, setdiff(filteredStarsoloTEs, plateletBarcodes)), ]

print("Summary of n counts")
print(summary(objTE_STARsolo$nCount_RNA))
print("Summary of n feature")
print(summary(objTE_STARsolo$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(objTE_STARsolo) %in% annotation_stellarscope$stellarscopeID))
```

```
[1] "Summary of n counts"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  90.33  374.02  465.87  546.65  653.93 2485.34
[1] "Summary of n feature"
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   96.0   353.0   438.5   485.6   569.2  1668.0
[1] "Loci present in the annotation:"

TRUE
4955
```

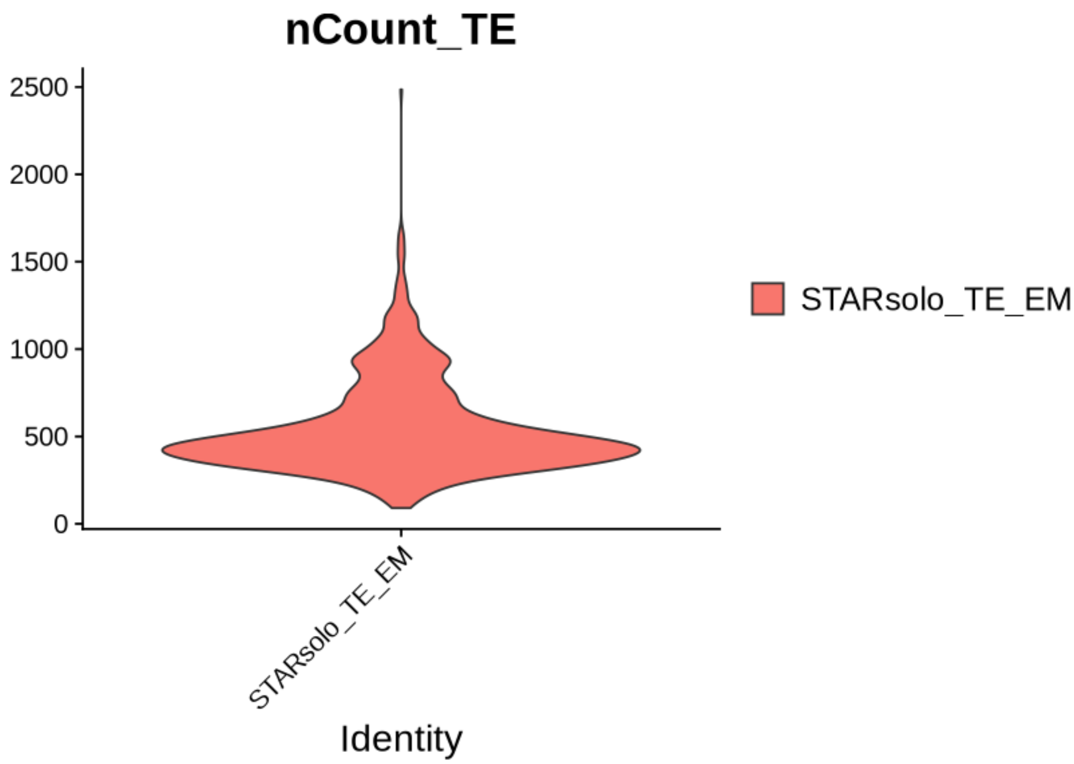
[18]:
```r
feature_metadata <- annotation_stellarscope[match(Features(objTE_STARsolo), annotation_stellarscope$stellarscopeID),]

head(feature_metadata)
```

A data.frame: 6 × 10

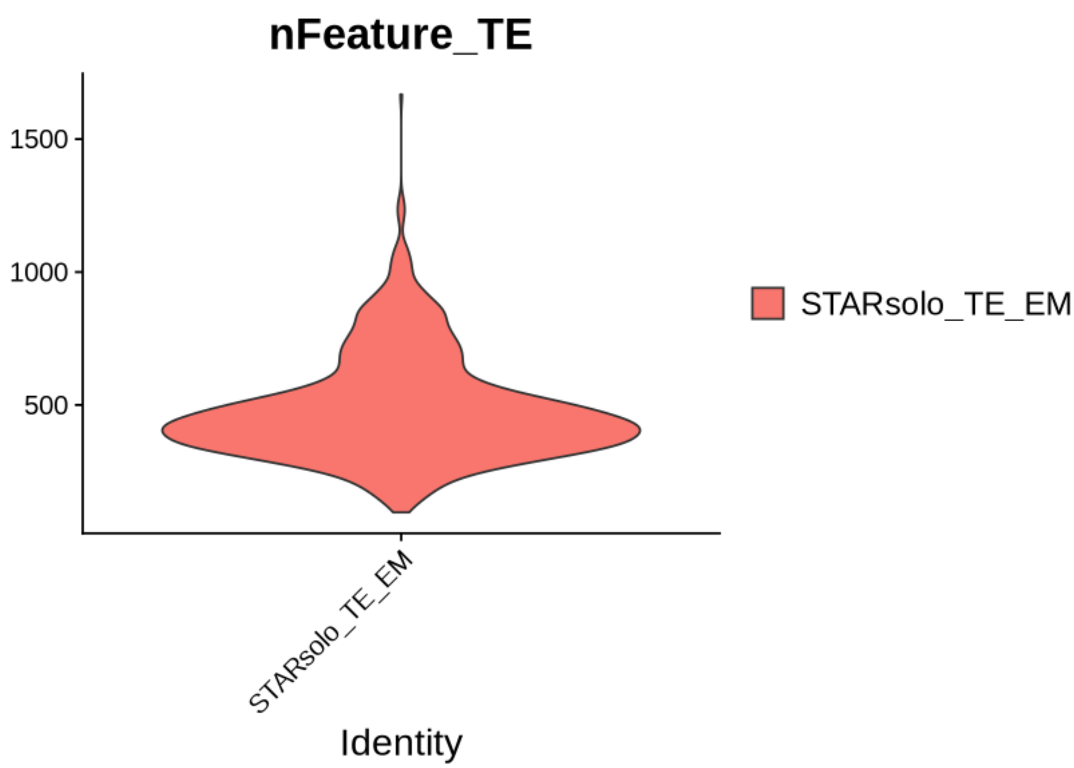
	chr	start	end	sw_score	strand	locusID	family	s
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<
269	chr1	43384732	43384958	1005	-	MIRb_dup9	MIR	M
584	chr1	150732607	150732908	2155	-	AluSz6_dup5	Alu	A
1173	chr1	164359	164695	1951	+	AluSq2_dup22	Alu	A
1657	chr1	522955	523291	1951	+	AluSq2_dup33	Alu	A
2217	chr1	958244	958549	2303	-	AluSx1_dup51	Alu	A
2281	chr1	1007747	1008060	2612	-	AluYa5_dup1	Alu	A

[19]:
```r
options(repr.plot.width=7, repr.plot.height=5)


objTE_STARsolo@meta.data$nCount_TE <- objTE_STARsolo@meta.data$nCount_RNA
objTE_STARsolo@meta.data$nFeature_TE <- objTE_STARsolo@meta.data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_STARsolo, features = c("nCount_TE"), ncol = 1,
        pt.size = 0) + theme(text=element_text(size=17))

ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_STARsolo.png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_STARsolo.pdf"), device='pdf')

VlnPlot(objTE_STARsolo, features = c("nFeature_TE"), ncol = 1,
         pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_STARsolo.png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_STARsolo.pdf"), device='pdf')
```

```
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Saving 7 x 7 in image
Saving 7 x 7 in image
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."
```

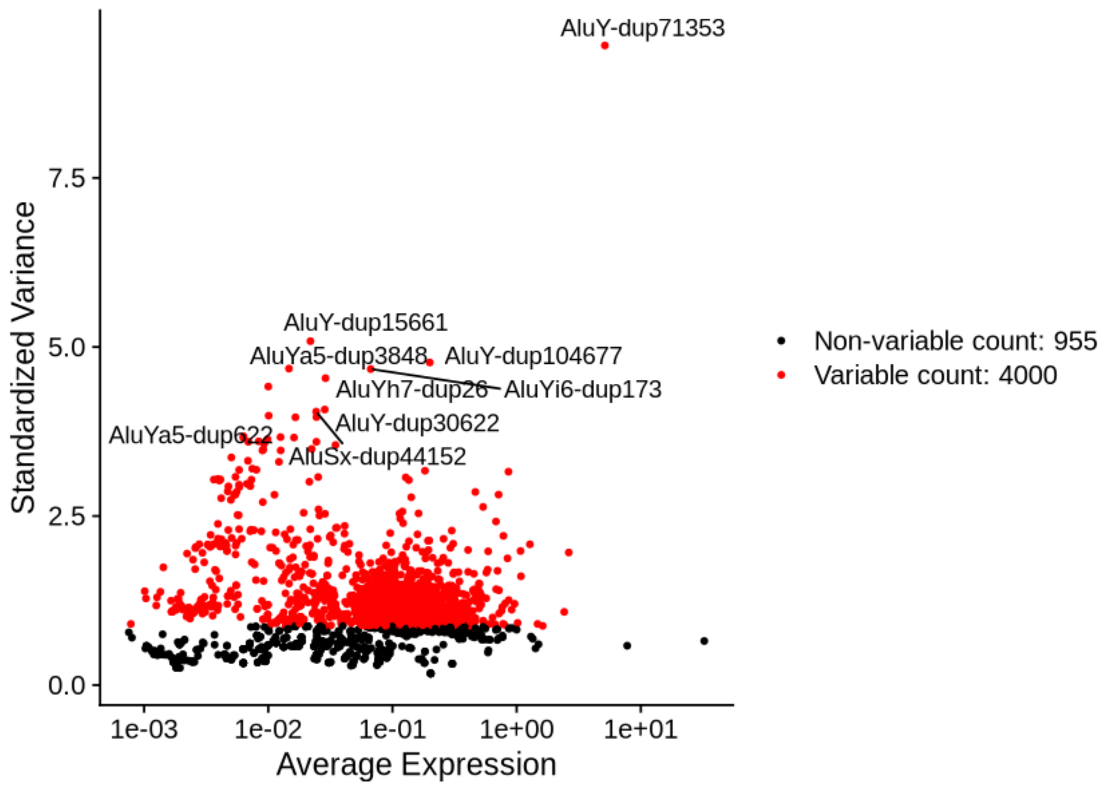



```
Saving 7 x 7 in image
Saving 7 x 7 in image
```



[20]:
```
objTE_STARsolo <- JoinLayers(objTE_STARsolo)

objTE_STARsolo <- NormalizeData(objTE_STARsolo, normalization.method =
 ↪"LogNormalize", scale.factor = 10000)

objTE_STARsolo <- FindVariableFeatures(objTE_STARsolo, selection.method =
 ↪"vst", nfeatures = 4000)

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_STARsolo), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_STARsolo)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

```
Normalizing layer: counts

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results
```



[21]:
```
gc()
all.genes <- rownames(objTE_STARsolo)
objTE_STARsolo <- ScaleData(objTE_STARsolo) # on hvgs
```

A matrix: 2 × 6 of type dbl

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	18215828	972.9	45600291	2435.4	29366755	1568.4
Vcells	105070802	801.7	232762587	1775.9	231728777	1768.0

```
Centering and scaling data matrix
```

[22]:
```
objTE_STARsolo <- RunPCA(objTE_STARsolo, features = VariableFeatures(object = 
  ↪objTE_STARsolo))

DimPlot(objTE_STARsolo, reduction = "pca") + NoLegend()

ElbowPlot(objTE_STARsolo)
```

```
PC_ 1
Positive:  AluSz-dup94020, AluSz-dup85172, AluSz6-dup5, AluSc5-dup6172,
L1ME3A-dup871, AluY-dup71353, Tigger1-dup5817, AluJb-dup126028, AluYa5-dup1820,
MER39-dup143
           AluSx1-dup84605, AluSq-dup11392, AluJb-dup42076, AluJb-dup110283,
```

L2a-dup163990, AluSz6-dup7653, AluSx-dup105425, AluJo-dup60604, MIR-dup162327, AluSp-dup32375
L2b-dup93969, AluSp-dup29416, AluSx1-dup105068, AluSz6-dup29860, AluJb-dup3312, AluJb-dup126027, AluSx1-dup33190, L1PA8-dup659, AluJb-dup123770, AluSx3-dup20249
Negative: AluSx-dup97345, AluSx-dup129237, AluSz-dup65970, MLT-int-dup273, L2b-dup1981, Harlequin-int-dup426, AluJb-dup73017, AluSx-dup34093, AluSz-dup65021, AluSx1-dup96779
AluSx1-dup122805, AluYf1-dup1634, AluSx1-dup56311, AluSz-dup75971, AluSp-dup3380, AluSp-dup36543, AluSz-dup76216, L1MA9-dup5209, L1M4-dup14436, AluSq2-dup15156
AluJo-dup55958, AluJb-dup29720, L1MA7-dup6321, AluSp-dup17111, AluSx1-dup65945, AluSq2-dup37910, MLT1A-dup5928, AluJb-dup43713, AluSx1-dup39416, L1PA15-dup2196
PC_ 2
Positive: AluY-dup30607, AluY-dup30565, AluY-dup106751, AluY-dup106657, AluY-dup71353, AluYe5-dup191, Tigger1-dup5817, L1ME3A-dup8534, AluSz6-dup29860, AluSz6-dup5
AluSq2-dup8938, AluSx1-dup107728, AluSz-dup85172, AluSp-dup32375, MER39-dup143, AluSz-dup102545, L2a-dup163990, AluSg-dup22148, AluSg-dup5590, AluJo-dup22815
AluY-dup87975, AluSx1-dup54140, L1ME3Cz-dup5742, AluY-dup32754, AluY-dup104677, AluSg-dup23720, AluJo-dup47192, AluJo-dup72542, AluJb-dup70809, AluJb-dup3312
Negative: AluY-dup15163, AluSx-dup74302, AluSg-dup28111, AluJo-dup73332, AluSc8-dup19219, AluSx1-dup46526, AluSx1-dup109068, MLT1A0-dup19532, AluSx4-dup3994, AluY-dup98819
FLAM-A-dup2287, AluSx1-dup88383, AluSx4-dup1648, AluSq-dup8331, AluSx-dup91418, AluSx1-dup18933, AluY-dup1142, AluSx1-dup46199, AluSg-dup34300, AluY-dup48440
AluSx-dup112107, L1PA7-dup2614, AluSc5-dup2592, AluY-dup44374, AluSx1-dup63529, AluY-dup101321, AluSx-dup101543, AluSp-dup22634, AluSz-dup58609, AluY-dup102456
PC_ 3
Positive: AluSx-dup129237, AluSx-dup97345, AluSx3-dup18906, AluSx1-dup96779, AluSg4-dup4752, AluSx1-dup122805, AluSz-dup65970, MLT-int-dup273, AluSp-dup36543, Harlequin-int-dup426
AluJo-dup55958, MIRc-dup69564, L2b-dup1981, AluSg-dup2258, AluSp-dup21270, AluSg-dup15181, AluSx1-dup65945, L1MA7-dup6321, AluSx-dup105425, AluSz-dup76216
AluSc8-dup1275, AluSx-dup100843, AluSz-dup65021, AluSx1-dup105068, AluSz6-dup42581, AluJb-dup73017, AluSx1-dup38257, MIRc-dup98292, AluSz-dup58855, AluSp-dup3380
Negative: AluSc5-dup6699, L1MC5a-dup13849, AluY-dup104962, AluSx1-dup105649, AluJr-dup84944, MLT1A0-dup938, AluJb-dup43713, AluJr-dup84945, MIRc-dup14097, AluSq2-dup14424
AluJb-dup86231, AluSq-dup1994, AluSx-dup6825, AluSx1-dup13093, AluSz6-dup32784, AluY-dup46259, AluY-dup53734, AluSc-dup20728, L1MB5-dup1608,

AluJb-dup76256
AluSx1-dup33189, L1PA4-dup2559, AluSq-dup1899, L1M5-dup40421, AluY-dup12183, AluSz6-dup21511, AluJr-dup26147, AluSx1-dup13094, AluJb-dup104665, AluSq2-dup7220

PC_ 4

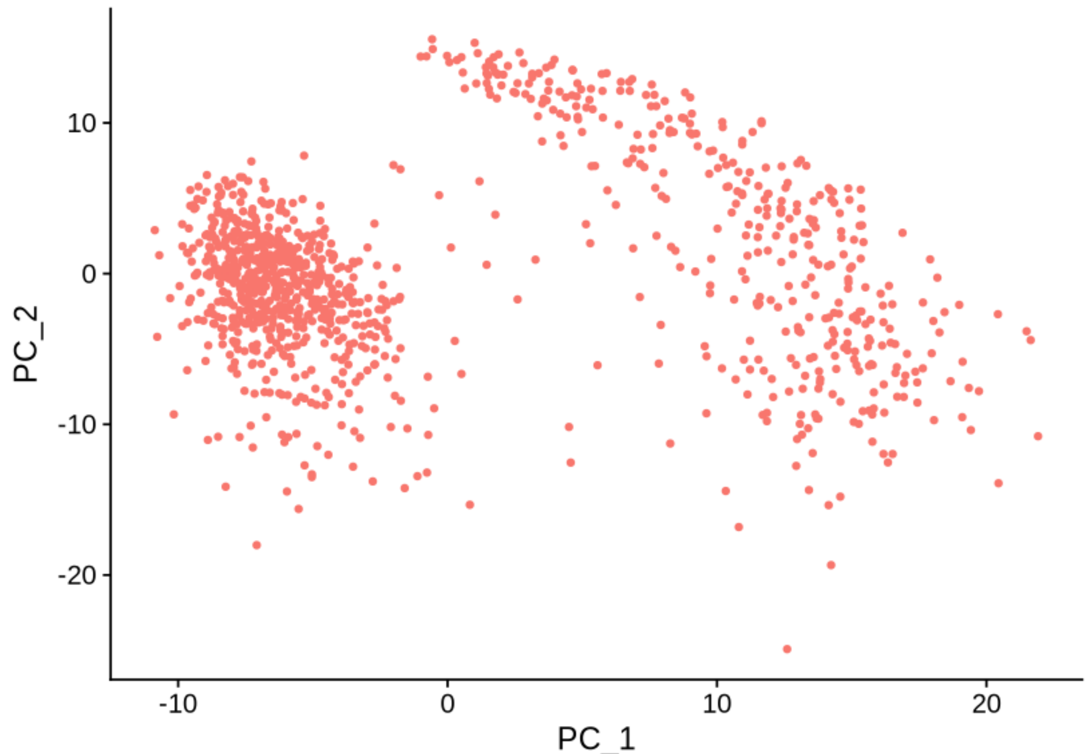
Positive: AluY-dup106751, AluY-dup30565, AluY-dup30607, AluY-dup106657, AluY-dup71353, AluJr-dup72756, AluSz-dup73882, AluSx1-dup30337, AluSz-dup73886, AluY-dup82338
AluSx1-dup109068, AluSx1-dup73735, AluYe5-dup184, L1PA10-dup1255, AluSp-dup32315, AluSp-dup22634, AluSx-dup103749, AluJb-dup3312, AluSz-dup87950, MIR-dup162327
AluJb-dup36574, AluSx-dup3431, AluYk4-dup915, AluSp-dup32314, AluSz6-dup2798, AluSq-dup8331, MER11C-dup315, AluJb-dup8078, AluSq2-dup58238, AluSx1-dup83623

Negative: AluJr-dup82212, AluSx1-dup51724, AluY-dup13774, AluSg-dup30170, AluSz6-dup22130, AluSz-dup84914, L2a-dup163990, AluSx3-dup20134, AluSc-dup32567, MER41A-dup2380
L1MA9-dup11533, AluSz6-dup44790, AluSx1-dup84605, AluSx1-dup65945, L1PA4-dup5055, L3-dup40489, MER5A-dup24456, AluSx-dup97345, AluSx1-dup70012, AluJb-dup110283
AluSx-dup129237, AluSx1-dup84603, AluSx-dup44855, MIRc-dup98292, AluJb-dup45602, AluSp-dup12283, AluY-dup65449, L1MC2-dup387, AluSx1-dup35031, AluSz-dup65968

PC_ 5

Positive: L2a-dup163990, AluSx1-dup30337, AluJr-dup82212, AluSq2-dup15156, AluSc-dup32567, AluSx1-dup84605, AluSx3-dup20134, AluSz6-dup44790, AluJb-dup36574, AluSq2-dup8706
AluSx-dup33084, L1PA15-dup2196, AluSg7-dup5445, AluSx-dup99692, AluSx-dup32527, AluSx-dup69559, AluSq-dup2132, AluSz-dup18708, HAL1-dup18700, AluSq2-dup38744
AluSx1-dup27642, L2a-dup163989, AluSx1-dup56311, AluSc8-dup6033, AluJr-dup59248, AluSx1-dup84603, L1MA9-dup11533, AluSz-dup28610, MER41A-dup2380, AluY-dup27178

Negative: AluSx-dup129237, AluSx-dup97345, AluSx1-dup65945, Tigger3b-dup268, MIRc-dup98292, AluSz-dup65968, AluSx1-dup73638, AluSx1-dup122805, AluSx1-dup96779, MLT-int-dup273
AluSq2-dup24157, AluSx-dup44855, AluSx-dup5374, AluSz-dup65970, AluSz-dup47729, AluJb-dup5937, AluSx1-dup98432, AluY-dup709, AluSp-dup31765, AluJr-dup17451
AluSx-dup6849, L1MB7-dup15579, AluY-dup87172, L1MA7-dup6321, AluSx1-dup88436, AluY-dup1142, AluSp-dup48583, AluY-dup69173, AluSx-dup108887, AluSx-dup3267



10
0
-10
-20
PC_2
-10
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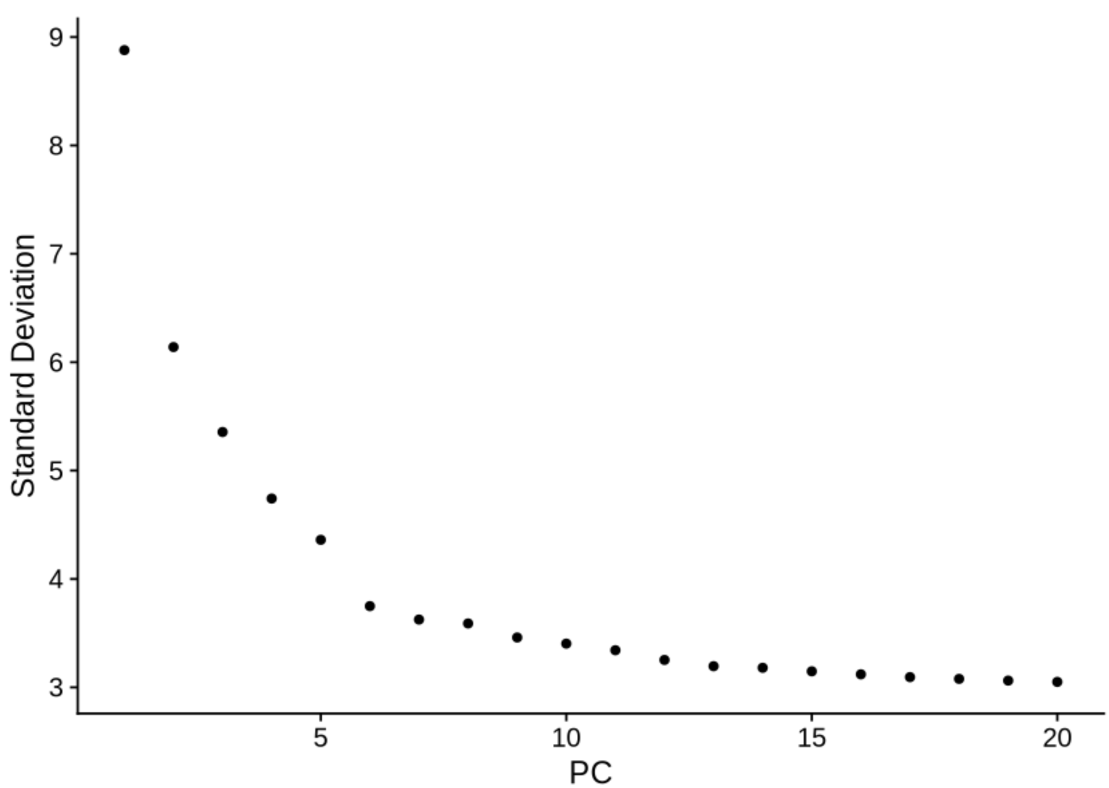
PC_1



9
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3
Standard Deviation
5
10
15
20
PC

[23]:
```
objTE_STARsolo <- FindNeighbors(objTE_STARsolo, dims = 1:13, k.param = 20)
objTE_STARsolo <- FindClusters(objTE_STARsolo, resolution = 0.5)
objTE_STARsolo <- RunUMAP(objTE_STARsolo, dims = 1:13)
DimPlot(objTE_STARsolo, reduction = "umap")
```

```
Computing nearest neighbor graph


Computing SNN


Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 1056
Number of edges: 37997

Running Louvain algorithm...
Maximum modularity in 10 random starts: 0.8497
Number of communities: 7
Elapsed time: 0 seconds

15:34:26 UMAP embedding parameters a = 0.9922 b = 1.112

15:34:26 Read 1056 rows and found 13 numeric columns

15:34:26 Using Annoy for neighbor search, n_neighbors = 30

15:34:26 Building Annoy index with metric = cosine, n_trees = 50

0%   10   20   30   40   50   60   70   80   90   100%

[----|----|----|----|----|----|----|----|----|----|

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15:34:26 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpnxZt1j/fileacfd239750b20

15:34:26 Searching Annoy index using 1 thread, search_k = 3000

15:34:26 Annoy recall = 100%

15:34:28 Commencing smooth kNN distance calibration using 1 thread
 with target n_neighbors = 30
```
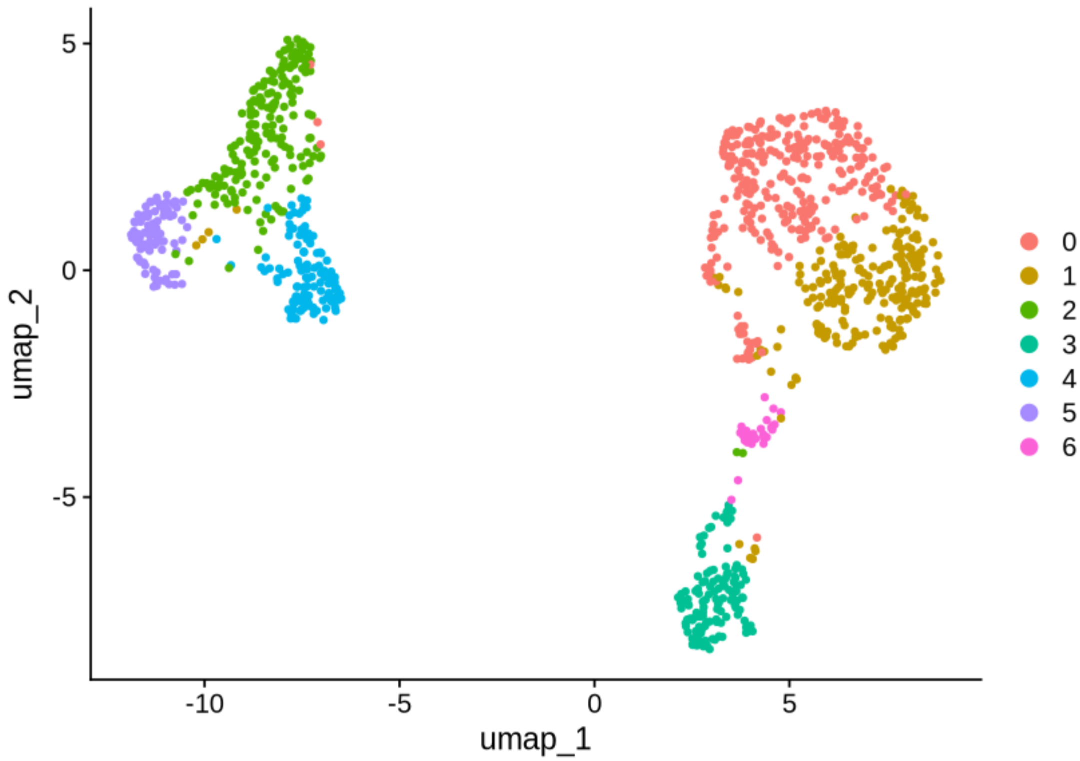

```
15:34:32 Initializing from normalized Laplacian + noise (using RSpectra)

15:34:32 Commencing optimization for 500 epochs, with 41350 positive edges

15:34:32 Using rng type: pcg

15:34:36 Optimization finished
```

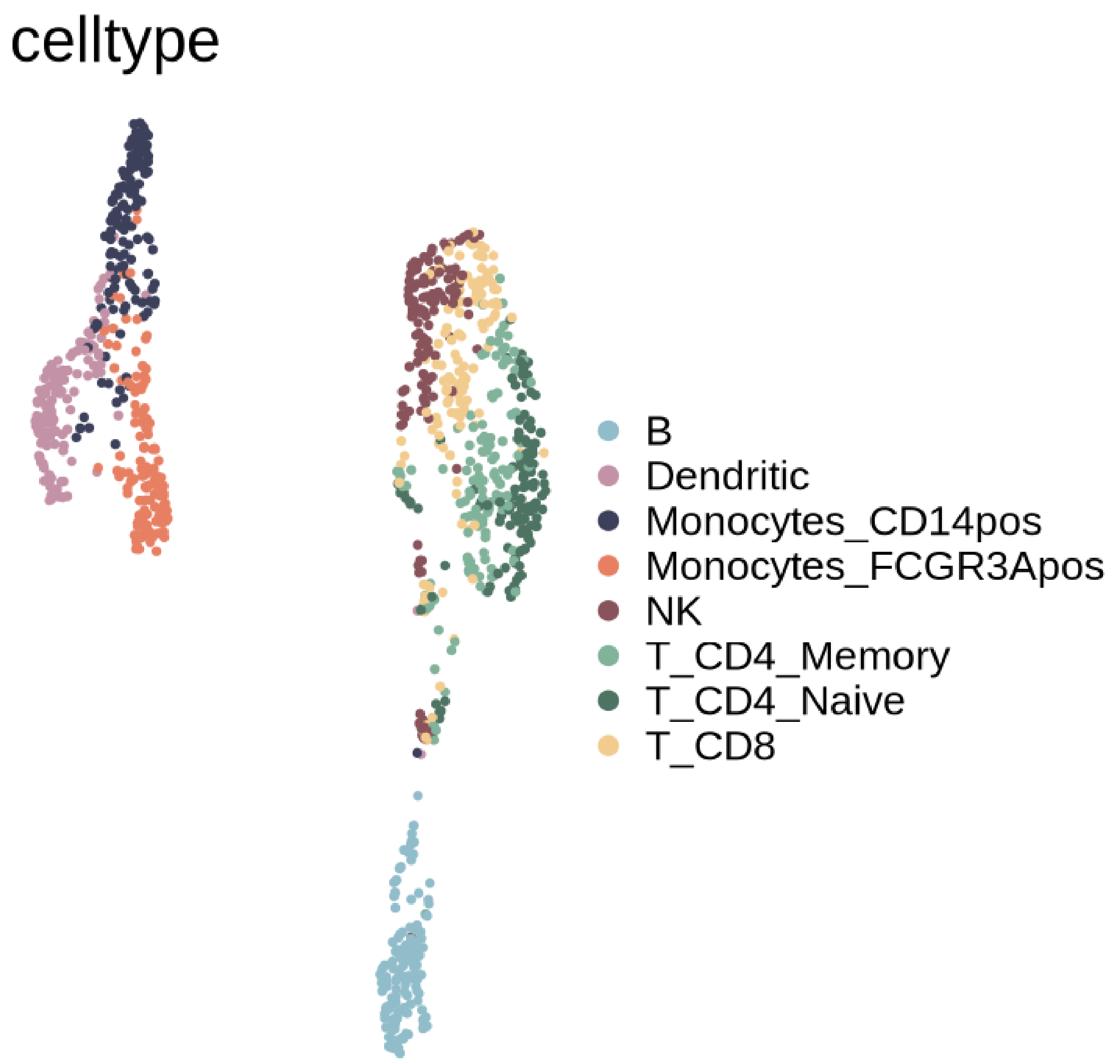


[24]:
```
# celltypes <- read.table("/mnt/TEdata/TEbenchmarking/data/whitelists/
↪celltype_annotation_sub_min.tsv")
# table(Cells(objTE_STARsolo) %in% celltypes$V1)

objTE_STARsolo$celltype <- celltypes$V2[match(Cells(objTE_STARsolo),␣
↪celltypes$V1)]
table(objTE_STARsolo$celltype)
```

```
                  B           Dendritic    Monocytes_CD14pos Monocytes_FCGR3Apos
                132                 132                  132                 132
                 NK         T_CD4_Memory          T_CD4_Naive               T_CD8
```

132 132 132 132

[25]:
```
options(repr.plot.width=6, repr.plot.height=6)

DimPlot(objTE_STARsolo, reduction = "umap", group.by = "celltype",
        cols=PBMCcelltypeColors,
        shuffle=T, pt.size = 1) +
  theme_void() +
  theme(text=element_text(size=20))
ggsave(paste0("figures_",dataset_id,"/
↪umap_TEs_locus_celltypes_STARsolo_noplatelet.pdf"), device = "pdf", width=9.
↪5, height=7)
```



[ ]:

```

```

[26]:

```
saveRDS(objTE_STARsolo, paste0("data_",dataset_id,"/
 ↪STARsolo_",dataset_id,"_seuratObj_noPlatelet.RDS"))
```

[ ]:

```

```